

# Conflict-Aware Cross-View Disaster Triage Across Spatial Observation Regimes [experiments]

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## ABSTRACT

Cross-view disaster assessment combines ground-level and overhead imagery to infer building damage. Aggregate F1 scores, however, hide the cases where fusion matters most: examples for which the single-view predictors disagree. This study introduces a Conflict-Aware Evaluation protocol (CAE) for paired-view disaster triage. CAE reports overall F1, conflict-subset F1, and the conflict gain of cross-view fusion over the stronger single-view baseline. We apply CAE to two paired disaster settings: an Eaton/Altadena wildfire dataset with property-centered ground imagery and the Hurricane Ian CVDIsaster benchmark with panoramic ground imagery. The results show that cross-view fusion is most informative on disagreement cases, but the dominant single-view arbitrator depends on the observation regime and the split definition. On the wildfire sensitivity split, cross-view fusion improves conflict F1 by 0.0388 over the best single-view baseline. On a leakage-controlled hurricane endpoint split, cross-view fusion no longer dominates street-only inference on conflict cases, which provides an important boundary condition. We further report threshold, seed, backbone, spatial, training, and routing checks that test whether this conflict-centered view persists beyond one split or one model instance. The evidence supports the central claim: paired-view disaster models should be evaluated as conflict resolvers, not only as average-case classifiers.

## KEYWORDS

disaster assessment, cross-view learning, multimodal fusion, GeoAI, conflict-aware evaluation

## 1 INTRODUCTION

Post-disaster building assessment increasingly uses several visual viewpoints. Overhead imagery gives broad spatial coverage, while ground imagery can expose facade condition, debris, and local structural cues. Cross-view models seek to combine these signals, but prior evaluation often reduces their value to one aggregate test metric. That view is too coarse for operational disaster triage. Many buildings are easy in both views. The harder and more consequential cases are those where the two views provide competing evidence.

This paper studies cross-view fusion as a conflict-resolution problem. We focus on paired examples that contain a ground image, an overhead patch, and a binary damage label. Instead of asking only whether paired-view fusion improves average F1, we ask when fusion helps after the two single-view models disagree. That framing matters for geographic AI because observation conditions are not interchangeable. A close property inspection photograph and a panoramic street scene both count as ground imagery, but they

encode different relationships between observer, target building, and surrounding landscape.

The study sits between several lines of disaster-vision and geospatial evaluation work. Satellite damage benchmarks such as xBD have made overhead-only building assessment a standard task [2], and later fusion models improve satellite-based damage assessment from multi-scale visual evidence [8]. Ground and street-view studies show that close-range imagery can capture damage cues that overhead imagery misses [9, 11, 12]. CVDIsaster is the closest paired-view benchmark for Hurricane Ian [4]. Cross-view geolocalization work has also established the broader term for matching ground and aerial views [7]; here the goal is not localization but damage triage under evidence conflict. CAE is related to selective prediction and disagreement-aware evaluation [1], but it does not claim a new risk-coverage estimator. Its contribution is a spatially grounded reporting unit for paired disaster views: the subset where the two observation regimes disagree, the single-view direction of correction, and the gain or loss from fusion. This complements spatial evaluation concerns about leakage and split design [6]. Recent generative work also studies how post-disaster ground views may be synthesized from satellite imagery when field observations are sparse [10].

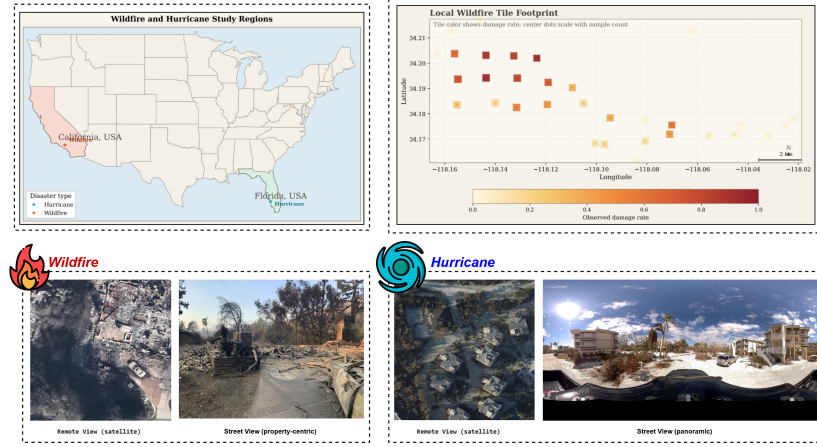
This study makes three contributions. First, it defines Conflict-Aware Evaluation (CAE), a three-metric protocol that reports full-test performance and conflict-subset behavior together. Second, it applies CAE to wildfire and hurricane paired-view triage, while treating the hurricane grouped-clean split as the conservative endpoint control. Third, it shows that the disagreement signal remains useful beyond reporting: it supports threshold-sensitivity checks, tile-level conflict-density mapping, conflict-weighted training, and an adaptive inference cascade. The analysis deliberately avoids a causal claim that disaster type alone determines which view should dominate. The evidence is better read as a regime-sensitive experiment: cross-view fusion helps most when single-view evidence conflicts, and the source of that help depends on how the ground view observes the target.

## 2 DATA AND PROTOCOL

### 2.1 Paired-View Triage Settings

Let  $\mathcal{D} = \{(x_i^{(s)}, x_i^{(r)}, y_i)\}_{i=1}^N$  be a paired disaster triage dataset, where  $x_i^{(s)}$  is a street or ground image,  $x_i^{(r)}$  is an overhead patch, and  $y_i \in \{0, 1\}$  is the binary damage label. We compare three modes: *street-only* (S), *remote-only* (R), and *cross-view* (CV). The

(a) Spatial Observation Regimes and Geographic Context



**Figure 1: Study design and spatial observation regimes.** The wildfire setting uses property-centered inspection imagery paired with overhead patches, whereas the hurricane setting uses panoramic ground imagery paired with overhead patches. This visual difference motivates the regime-aware interpretation tested by CAE.

**Table 1: Evaluation settings and review controls.**

| Setting     | Positive class | Ground regime | Role            |
|-------------|----------------|---------------|-----------------|
| WF sens.    | affected+      | property      | main CAE        |
| H end.      | severe         | panoramic     | diagnostic      |
| H mod.+sev. | moderate+      | panoramic     | sensitivity     |
| H clean     | severe         | panoramic     | leakage control |

cross-view model uses late fusion of ResNet18 features [3], including each embedding, their absolute difference, and their elementwise product. This simple fusion design keeps the evaluation focused on conflict behavior rather than architectural novelty.

The wildfire setting uses an Eaton/Altadena paired dataset assembled for this project. Its ground imagery is property-centered inspection imagery paired with overhead patches. This observation regime often places the target building near the visual center of the ground image. The hurricane setting uses the Hurricane Ian CVD disaster benchmark [4], which pairs panoramic ground imagery with very high resolution overhead imagery. This observation regime often includes wider street context, adjacent buildings, and weaker target alignment. The comparison therefore mixes disaster type, geography, source, and collection protocol; those factors limit causal interpretation and motivate the conservative language used below.

Table 1 records the experimental settings and the control role of each split. This is important because the paper should not choose a split after observing which one makes the conflict story look strongest. The wildfire sensitivity split is used because it gives a larger and more informative disagreement subset. The hurricane grouped-clean endpoint split is included because it controls object-level leakage in the original endpoint partition.

## 2.2 Conflict-Aware Evaluation

For each example, let  $p_i^{(s)}$  and  $p_i^{(r)}$  denote the positive-class probabilities predicted by the street-only and remote-only models. CAE

defines the conflict subset

$$C_\tau = \{i : |p_i^{(s)} - p_i^{(r)}| > \tau\}, \quad (1)$$

with  $\tau = 0.1$  in the main analysis. For each model  $m$ , CAE reports standard overall F1 and conflict F1 on  $C_\tau$ . It then computes the cross-view conflict gain

$$\Delta_{\text{view}} = F1(\text{crossview}; C_\tau) - \max\{F1(\text{street}; C_\tau), F1(\text{remote}; C_\tau)\}. \quad (2)$$

This protocol exposes whether paired-view fusion resolves disagreement cases rather than only improving easy average cases.

We also decompose conflicts by correction direction.  $S \rightarrow O$  denotes cases where the street-only model is correct and the remote-only model is wrong;  $O \rightarrow S$  denotes the reverse. These fractions identify the stronger single-view arbitrator inside the conflict subset. Because a prior hurricane endpoint split had object-level leakage, we report the grouped-clean endpoint split as a required control rather than as an optional robustness run.

## 2.3 Implementation Details

All modes use the same triage implementation. Images are resized to 224 by 224 pixels and normalized for the selected backbone. The main runs use ImageNet pretrained ResNet18 encoders. Each view embedding is projected to 256 dimensions; the cross-view classifier receives the street embedding, overhead embedding, absolute difference, and elementwise product. Models are optimized with AdamW using learning rate  $10^{-4}$  for 10 epochs, and the checkpoint with the best validation F1 is evaluated on the held-out test split. This keeps the comparison between street-only, remote-only, and cross-view models tied to the same training protocol.

## 3 RESULTS

### 3.1 CAE Summary

Table 2 reports the core CAE results. The wildfire sensitivity split has the clearest positive conflict gain: cross-view fusion reaches

**Table 2: Conflict-Aware Evaluation summary. Conflict F1 uses  $C_\tau$  with  $\tau = 0.1$ . WF is wildfire; H is hurricane. Gain is reported only for cross-view rows.**

| Setting     | M  | All    | C-F1   | Gain    | $n_c$ |
|-------------|----|--------|--------|---------|-------|
| WF sens.    | S  | 0.9392 | 0.6230 | –       | 435   |
| WF sens.    | R  | 0.9238 | 0.5237 | –       | 435   |
| WF sens.    | CV | 0.9473 | 0.6618 | +0.0388 | 435   |
| H end.      | S  | 0.9055 | 0.6842 | –       | 32    |
| H end.      | R  | 0.8969 | 0.5806 | –       | 32    |
| H end.      | CV | 0.9192 | 0.7647 | +0.0805 | 32    |
| H mod.+sev. | S  | 0.8738 | 0.7324 | –       | 128   |
| H mod.+sev. | R  | 0.8873 | 0.7742 | –       | 128   |
| H mod.+sev. | CV | 0.8915 | 0.7867 | +0.0125 | 128   |
| H clean     | S  | 0.9016 | 0.8101 | –       | 80    |
| H clean     | R  | 0.8385 | 0.6526 | –       | 80    |
| H clean     | CV | 0.9008 | 0.7949 | -0.0153 | 80    |

**Table 3: Directional conflict decomposition at  $\tau = 0.1$ .  $S \rightarrow O$  is street-correct and remote-wrong;  $O \rightarrow S$  is the reverse.**

| Split       | $n_c$ | $S \rightarrow O$ | $O \rightarrow S$ | Stronger view |
|-------------|-------|-------------------|-------------------|---------------|
| WF sens.    | 435   | 0.2713            | 0.1885            | street        |
| H end.      | 32    | 0.3750            | 0.3438            | near-tie      |
| H mod.+sev. | 128   | 0.1094            | 0.1328            | remote        |

0.6618 conflict F1, which is 0.0388 above the better single-view baseline. The original hurricane endpoint split also shows a large gain, but its small conflict subset ( $n = 32$ ) and leakage history make it diagnostic rather than primary evidence. The broader hurricane moderate-plus-severe split keeps a smaller positive gain. The grouped-clean endpoint split provides the most important boundary condition: cross-view fusion remains close to street-only overall, but it no longer dominates street-only on the conflict subset.

The directional decomposition sharpens this interpretation. On wildfire,  $S \rightarrow O$  conflicts are more common than  $O \rightarrow S$  conflicts (0.2713 versus 0.1885), so the ground image is the stronger single-view arbitrator. On the hurricane endpoint split, the two directions are near tied (0.3750 versus 0.3438). On the broader hurricane split, the ordering reverses (0.1094 versus 0.1328), making the overhead view the stronger single-view arbitrator. The stable claim is therefore not that one disaster always favors one modality. It is that conflict behavior changes with the observation regime and split definition.

### 3.2 Robustness Signals

The conflict effect is not only a single threshold artifact. A separate threshold sweep at  $\tau = \{0.1, 0.3, 0.5\}$  keeps CV strongest on wildfire conflict accuracy (0.8349, 0.7821, 0.7538) and hurricane conflict accuracy (0.7097, 0.6400, 0.6190). Under the reporting threshold  $\tau = 0.1$ , CV accuracy on conflict cases is 0.7885 for the wildfire sensitivity split, compared with 0.7356 for S and 0.6529 for R. On the hurricane moderate-plus-severe split, the corresponding values are 0.7500, 0.7031, and 0.7266. Label-independence permutation tests are below  $10^{-4}$  in both settings. Multi-seed validation also supports the base ordering: wildfire CV reaches  $0.9689 \pm 0.0011$  validation F1, while S and R reach  $0.9657 \pm 0.0001$  and  $0.9654 \pm 0.0015$ . Hurricane CV reaches  $0.9455 \pm 0.0063$ , compared with  $0.9433 \pm 0.0012$  and  $0.8959 \pm 0.0065$ . Backbone checks add a caution: ResNet18

**Table 4: Auxiliary analyses and operational extensions.**

| Analysis          | Primary signal                          | Scope    |
|-------------------|-----------------------------------------|----------|
| Threshold sweep   | CV remains best at 3 $\tau$ values      | WF/H     |
| Conflict density  | $r = 0.5123$ , uncorrected $p = 0.0063$ | WF tiles |
| ConflictFocalLoss | best conflict F1 = 0.7208               | WF split |
| Adaptive cascade  | F1 = 0.9498, 9.5% routed                | WF split |

and ResNet50 remain strong, while generic CLIP and DINOv2 features underperform the supervised CNN baselines in this disaster-specific triage setting.

### 3.3 Spatial and Training Extensions

CAE also produces useful secondary signals. For the wildfire dataset, we aggregate  $|p_i^{(s)} - p_i^{(r)}|$  by overhead tile and compare the resulting conflict density with observed tile damage rate. Across 27 tiles, Spearman  $r = 0.5123$  and  $p = 0.0063$  before spatial-autocorrelation correction. This exploratory result suggests that cross-view disagreement can act as an unsupervised spatial damage proxy in the property-centered wildfire setting. We do not yet claim the same result for hurricane because the benchmark does not provide equivalent tile metadata in this repository.

We also test ConflictFocalLoss, a conflict-weighted binary loss related in spirit to focal reweighting [5]. For each cross-view example, the loss multiplies binary cross entropy by  $(1 + c_i)$ , where  $c_i$  is a feature-space disagreement score based on the cosine distance between the two view embeddings;  $c_i$  is computed with detached embeddings, so the weight does not back-propagate through the disagreement estimate. On the wildfire sensitivity split, a small gamma sweep shows that moderate conflict weighting is not a single-point accident: overall F1 is 0.9514, 0.9507, 0.9520, and 0.9470 for  $\gamma = 0.1, 0.25, 0.5, 1.0$ , compared with 0.9473 for standard CV. The best setting also improves conflict F1 from 0.6618 to 0.7208. This is a method extension validated in one setting, not a universal training recipe.

## 4 DISCUSSION

The experiment supports CAE as a useful evaluation protocol for paired-view disaster triage. Aggregate F1 alone would make the models look similar in some settings. CAE reveals how they behave on disagreement cases and shows that cross-view value can shrink, grow, or even become negative relative to the stronger single-view model after leakage control.

### 4.1 Evidence Scope and Data Release

The reproducible unit is the CAE computation: given held-out predictions from S, R, and CV models, CAE can be computed without changing the model architecture. This makes the protocol portable to other paired-view disaster datasets and to stronger fusion models, including attention-based, transformer-based, and generative-view designs. The current results support CAE as an empirical evaluation frame rather than a claim that one fusion architecture is universally optimal.

Two controls keep the contribution focused. First, the hurricane endpoint result is interpreted through the grouped-clean split. The

original endpoint split remains useful for benchmark continuity, but the leakage-controlled split sets the conservative boundary. Second, the wildfire dataset should be released with a data card covering split counts, class balance, image sources, collection timing, annotation rules, quality control, and access constraints. That documentation will make the contributed wildfire setting usable as a paired-view benchmark rather than only a private case study.

The spatial-map result has a similar boundary. The tile-level correlation is useful because it turns disagreement into a GIS-native damage signal, but the current statistic uses only 27 wildfire tiles. A final version should either report a spatial-autocorrelation check or explicitly describe the Spearman result as an exploratory spatial association. The same conservative stance applies to ConflictFocalLoss. Its wildfire gain is promising, but the loss should be compared against standard focal loss, class-balanced loss, and simple probability averaging before being framed as a standalone method contribution.

Several boundaries remain. The wildfire and hurricane settings differ by more than observation regime, so the results support a regime-sensitive interpretation rather than a causal proof. The hurricane endpoint conflict subset is small, so its large positive gain is diagnostic. ConflictFocalLoss and the tile-level spatial proxy are strongest on the wildfire setting and motivate broader validation. These boundaries do not weaken the core contribution: CAE shows when paired views resolve evidence conflict and when a single view remains sufficient.

## 4.2 Operational Implications

The practical use of CAE is a routing view of disaster triage. If street-only and remote-only predictions agree with high confidence, a response system can often avoid the cost of paired-view inference and use the cheaper agreement decision. If the two predictions disagree, the case becomes a candidate for cross-view fusion or human review. This is the logic behind the adaptive cascade result in Table 4: only 9.5% of wildfire samples are routed to the cross-view stage while preserving the cross-view performance level. The exact routing threshold should be tuned for each deployment, but the principle is simple: cross-view reasoning is most valuable when the evidence is already in tension.

The same view can guide dataset design. If a future benchmark wants to test observation-regime effects directly, it should collect property-centered and panoramic ground imagery for the same disaster and the same target buildings. That design would remove many of the confounders in the current comparison and would let CAE test whether target alignment, rather than hazard type or data source, drives the dominant-view switch. In this sense, the study offers an empirical warning and an evaluation protocol: average F1 is not enough for multimodal disaster assessment when the views disagree.

CAE can also support audit logs for triage escalation: the two single-view scores, conflict threshold, selected route, and final decision can be stored for each building. These records help target new

ground photos where overhead imagery is uncertain and prior conflicts show that ground evidence often corrects remote-only mistakes. The same log can evaluate generated satellite-to-street evidence by asking whether a synthesized view resolves conflict or reinforces the overhead model's prior.

## 5 CONCLUSION

This work reframes paired-view disaster triage as conflict-aware evaluation. Across wildfire and hurricane paired-view settings, cross-view fusion is most informative on disagreement cases, but the stronger single-view arbitrator changes with observation regime, split definition, and leakage control. The resulting lesson is practical: disaster benchmarks should report conflict-subset behavior alongside aggregate scores. Doing so makes multimodal fusion more falsifiable, more spatially interpretable, and more useful for deployment decisions.

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